

Operator Discovery Report: candidate_ranker_bottleneck_aware_2opt_moves

- Operator type: candidate_ranker.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic local ranking for 2opt-like moves: prioritize crossing/long-edge resolutions and cluster-spanning swaps, while penalizing nearest-neighbor trap moves; helps transfer on clustered and bottleneck instances w/
- Novelty classification: long_span_ranker.
- Rediscovery signature: candidate_ranker:delta=1.250,cross=1.250,span=0.600,trap=0.400.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- For each candidate move, compute a composite score from immediate delta, crossing removal, span, trap awareness, and local structure.
- Sort candidate moves by ascending composite score.
- Apply the first move that yields a genuine tour improvement.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: fb9684a7125b5c15.

Why It Should Help

- The operator changes which local-search moves are tried first, so it can improve search efficiency without rewriting the underlying solver.

Failure Cases

- Underperforms on grid_like_tsp with gap delta 0.054436.
- Does not transplant cleanly into random_restart_2opt.
- Does not transplant cleanly into clustered_local_search.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.

Transplant Results

- nearest_neighbor_2opt gap delta 0.001527, runtime delta 54.157271, distance-eval delta -100.571429.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 1836.116343, distance-eval delta 0.0.
- random_restart_2opt gap delta 0.006029, runtime delta -67.111943, distance-eval delta -42.285714.
- sparse_three_opt gap delta 0.0, runtime delta 4.127557, distance-eval delta 0.0.

- clustered_local_search gap delta 0.009035, runtime delta 50.913214, distance-eval delta -88.571429.

Instance-Family Results

- heldout_tsplib operator gap 0.236585 vs baseline 0.235058.
- clustered_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.161132 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.006293.
- Runtime inflation: 0.00507.
- Distance-evaluation inflation: 0.003718.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator -0.006293.
- simplified_delta_ranker mean gap 0.127067 and delta vs full operator -0.000543.
- shuffled_ranker mean gap 0.121771 and delta vs full operator -0.005839.